YEAR 10 MATHS · CHAPTER 9 · MIND MAP

Statistics

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1 Collecting representative data

A sample should represent the whole population. Random samples avoid bias.

Census: everyone. Sample: some

Random, systematic or stratified sampling

2 Data displays

Histograms for numerical data, column graphs for categories.

Shape: symmetric, positively skewed (tail to the right), negatively skewed, bimodal

3 Two-way tables

Count by two categories, then use the row and column totals.

25 of 40 students play sport; 15 of those are in Year 10

$\frac{15}{25} = 60\%$ of the sport players are in Year 10

4 Summary statistics

Centre: mean and median. Spread: range and IQR.

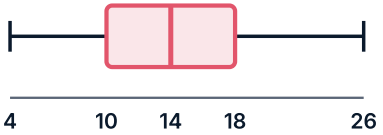
4, 6, 7, 9, 14: mean $\frac{40}{5} = 8$, median 7

An outlier pulls the mean, not the median.

5 Box plots and outliers

Outlier: below $Q_1 - 1.5 \times \text{IQR}$ or above $Q_3 + 1.5 \times \text{IQR}$.

$Q_1 = 10$, $Q_3 = 18$: IQR 8, upper fence $18 + 12 = 30$, so 34 is an outlier

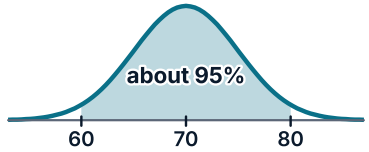


6 Standard deviation

Standard deviation σ measures spread around the mean. Small σ: data close to the mean.

Bell-shaped data: about 68% within 1σ, 95% within 2σ

Mean 70, $\sigma = 5$: about 95% between 60 and 80

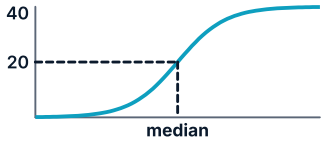


7 Cumulative frequency and percentiles

Cumulative frequency is a running total. Read percentiles off the graph (ogive).

Median = 50th percentile, Q_1 = 25th, Q_3 = 75th

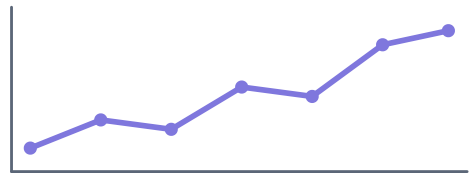
40 values: the median is halfway between the 20th and 21st



8 Time-series data

Data collected over time, with time on the x-axis.

Look for the trend (up or down) and any seasonal pattern



9 Bivariate data and scatter plots

Two numerical variables: explanatory on the x-axis, response on the y-axis.

Describe the direction (positive or negative), strength and form (linear or not)



Watch out: correlation doesn't prove that one causes the other.

10 Line of best fit

Draw it by eye with about the same number of points on each side.

$y = 2.5x + 10$ at $x = 8$: $y = 30$

Predicting inside the data (interpolation) is more reliable than outside it

